

QMWS Spring 2026: Introduction to Python for Data Analysis

Day One: Foundations

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Section 1

Installation

- A tool for configuring Python **environments**
 - More often than not, you need to install **packages** that allow you to code more easily/efficiently
 - Allows for version control
 - Issues can arise if environments are not used, for example:
 - If you install two packages that conflict with one another
 - If you install a package that requires a specific version of Python
 - Anaconda allows these issues to be isolated to a specific environment, rather than affecting your whole Python installation
 - Especially useful if you have more than one project going on

- Installation link: <https://www.anaconda.com/download>
- The simplest way to use Anaconda is via the **Navigator**
- To install:
 - 1 Make an account
 - 2 Install the full distribution via the “Graphical Installer”

- Installation link: <https://www.anaconda.com/download>
- On Mac:



Choose Your Download

Windows

Mac

Linux

Anaconda Distribution

Complete package with 8,000+ libraries, Jupyter, JupyterLab, and Spyder IDE. Everything you need for data science.

↓ [64-Bit \(Apple silicon\) Graphical Installer](#)

↓ [64-Bit \(Apple silicon\) Command Line Installer](#)

Miniconda

Minimal installer with just Python, Conda, and essential dependencies. Install only what you need.

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↓ [64-Bit \(Intel chip\) Graphical Installer](#)

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Setting Up a Conda Environment

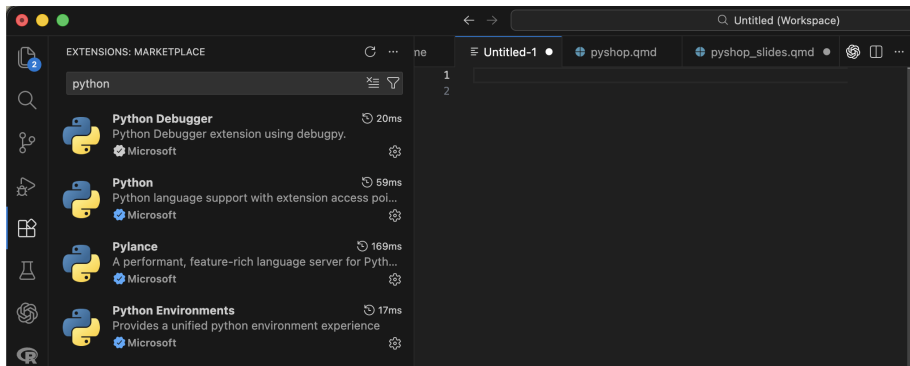
- See the presenter's shared screen

Visual Studio (VS) Code

- Installation link: <https://code.visualstudio.com/download>
- Integrated Development Environment (IDE)
- Can use it to code in many different languages
- Commonly used in both academia and industry

Visual Studio (VS) Code

- Once installed, go to the “Extensions” tab and search “Python”
- Install the below extensions:



- To point VS Code to your new Conda environment:
 - 1 Open the Command Palette
 - a. Ctrl + Shift + P on Windows
 - b. Command + Shift + P on MacOS
 - 2 Search “Python: Select Interpreter”
 - 3 Select the interpreter that corresponds to your new environment (the name of the environment will be in parentheses)

Section 2

The Basics

- In programming, a variable is a space in computer memory where values may be stored

Defining and Printing Variables

```
x = 67
```

```
print(x)
```

```
67
```

Arithmetic

```
a = 1
```

```
b = 2
```

```
print(a + b)      # * for multiplication, / for division
```

```
3
```

Note

- Use # to make a **comment**

Updating Variables

```
x = 10
```

```
x = 20
```

```
x = 15
```

```
print(x)
```

```
15
```

Variable Types

- ❶ `int` (integer; “whole number”)
 - less memory
 - 1 is an `int`
- ❷ `float` (real; “decimal number”)
 - 1.2 is a `float`
 - 1.0 is a `float`
- ❸ `str` (string; characters/words with no numerical meaning)
 - “Hello world!” is a string
 - “416-676-6767” is also a string
- ❹ `bool` (boolean; can be `True` or `False`; more on this later)
- ❺ `None`
 - A special type of variable that essentially means “nothing meaningful is stored”
 - Useful as a placeholder

Variable Types

Checking Type

```
x = 416
print(x, type(x))    # type() to check type

416 <class 'int'>
```

Converting Type

```
x = 416
x = str(x)           # str() to convert to string
print(x, type(x))

416 <class 'str'>
```

- Can also use `int()` and `float()` to convert to respective type

A Brief Note

- `print()`, `type()`, `str()`, etc. are all examples of **functions**
 - These are essentially pre-made chunks of code for you to use
 - More on this later!

What will the below code output?

```
x = "Variable"  
  
print(type(x))  
print(x)
```

Section 3

More Advanced Variable Types

- Multiple values can be stored in one variable. A list is one way to do this:

Defining Lists

```
x = [0, 1, 2, 3, 4]      # Define a list; square brackets  
print(type(x))
```

```
<class 'list'>
```

Lists: Indexing

```
x = [0,1,2,3,4]
```

- To extract one or more values from a list, you **index** the list
- Python uses **zero-based indexing**
 - This means that the first element of a list is index 0, the second element is index 1, ...
 - Element x is index $x - 1$

Indexing Lists

```
x = [0, 1, 2, 3, 4]
```

```
y = 2
```

```
print(x[0])      # Index first element of a list
```

```
print(x[-1])     # Index last element
```

```
print(x[y])      # Index yth element
```

```
0
```

```
4
```

```
2
```

Lists: Indexing

- You can also take more than one element, or a **slice**, of the list

Slicing Lists

```
x = [0, 1, 2, 3, 4]
```

```
print(x[0:2])    # First and second (but not third) elements
```

```
[0, 1]
```

! Important

- [inclusive:exclusive]
 - The index on the *left-hand* side of the slice is *included*, whereas the index on the *right-hand* side of the slice is *excluded*

Slicing Lists, continued

```
x = [0, 1, 2, 3, 4]
```

```
print(x[0:3]) # First three elements
```

```
print(x[0:1]) # Silly way to index just the first element
```

```
[0, 1, 2]
```

```
[0]
```

Lists: Length

- To find the length of a list:

len() function

```
x = [0, 1, 2, 3, 4]
```

```
print(len(x))    # The length of list x
```

```
# Another way to find the last element (not so silly!)
```

```
print(x[len(x) - 1])
```

5

4

Lists: Methods

- Lists have certain functions that can be applied to them, called **methods**
 - Methods are more general than this (e.g., string methods; more on these in a little bit!)

`.append()`

```
x = [0,1,2,3,4]
```

```
x.append(5)  
print(x)
```

```
[0, 1, 2, 3, 4, 5]
```

- Adds an entry to a list

`.extend()`

```
x = [0,1,2,3,4]
```

```
x.extend([6,7,8,9])
```

```
print(x)
```

```
[0, 1, 2, 3, 4, 6, 7, 8, 9]
```

- Extend a list by “glueing” it to another list

Difference Between `.append()` and `.extend()`

```
x = [0,1,2,4]
```

```
y = [5,6,7,8]
```

```
x.append(y)
```

```
print(x)
```

```
x = [0,1,2,4]    # Redefine x
```

```
x.extend(y)
```

```
print(x)
```

```
[0, 1, 2, 4, [5, 6, 7, 8]]
```

```
[0, 1, 2, 4, 5, 6, 7, 8]
```

Lists: Changing Values

Changing a Value in a List

```
x = [0, 1, 2, 3, 4]
```

```
x[0] = 9
```

```
print(x)
```

```
[9, 1, 2, 3, 4]
```

Tuples

- Like lists, tuples contain multiple values
- Unlike lists, tuples are **immutable**; they cannot be changed once created.
 - Lists are *mutable*

Defining a Tuple

```
x = ("a", "tuple")  # Round brackets
```

```
print(x[0], x[1])
```

```
print(type(x))
```

```
a tuple
```

```
<class 'tuple'>
```

Tuples

Changing Values... ERROR!

```
x = ("a", "tuple")
```

```
x[0] = "Not a"
```

```
TypeError: 'tuple' object does not support item assignment
```

TypeError

Traceback (most recent call last):

```
Cell In[18], line 3
```

```
1 x = ("a", "tuple")
```

```
2
```

```
----> 3 x[0] = "Not a"
```

```
TypeError: 'tuple' object does not support item assignment
```

- Tuples have special properties, one of which is below:

Unpacking Tuples

```
x, y = (6, 7)
```

```
print(x)
```

```
print(y)
```

```
6
```

```
7
```

Section 4

Strings

- Strings are a lot like lists in that they are **iterable**
 - For now, think of these (lists, strings) as a sequence of values

Strings are Like Lists!

```
cringe = "Hello world!"
```

```
print(cringe[0:5])  
print(len(cringe))
```

```
Hello  
12
```

- We can join two strings together through **concatenation**

Concatenation

```
a = "Hello"  
b = "World!"  
  
c = a + " " + b      # Note the space between words  
print(c)
```

```
Hello World!
```

- Sometimes you need to include quotations or other characters inside a string that would normally return an error
- This can be done through an **escape sequence**

ERROR! Why You Need Escape Sequences...

```
x = "This is a "string"
```

```
SyntaxError: invalid syntax (886282512.py, line 1)
```

```
Cell In[22], line 1
```

```
    x = "This is a "string"  
                ^
```

```
SyntaxError: invalid syntax
```

Escape Sequences

```
x = "This is a \"string\""
print(x)
```

This is a "string"

Note

- The \ indicates the start of an escape sequence

Strings

- Strings are also kind of like tuples in that they are **immutable**

String Immutability

```
cringe = "Hello world!"  
cringe[0:5] = "Goodbye"      # Error
```

TypeError: 'str' object does not support item assignment

TypeError Traceback (most recent call last):

Cell In[24], line 2

```
1 cringe = "Hello world!"  
----> 2 cringe[0:5] = "Goodbye"      # Error
```

TypeError: 'str' object does not support item assignment

- We can turn to string **methods** to “modify” strings when needed (they just create a new string “behind the scenes”)

```
.replace()
```

```
cringe = "Hello world!"
```

```
print(cringe.replace("Hello", "Goodbye"))
```

```
Goodbye world!
```

`.split()`

```
meme_list = "Harambe, 67, Jar Jar Binks, The Rizzler"
```

```
# Split on ", "; returns a list
```

```
new_list = meme_list.split(", ")
```

```
print(new_list)
```

```
print(new_list[0]) # Index the list
```

```
['Harambe', '67', 'Jar Jar Binks', 'The Rizzler']
```

```
Harambe
```

How might you index “The Rizzler”?

How might you index “The Rizzler”?

```
meme_list = "Harambe, 67, Jar Jar Binks, The Rizzler"  
new_list = meme_list.split(", ")
```

```
print(new_list[3])  
print(new_list[-1])  
print(new_list[len(new_list) - 1])
```

The Rizzler

The Rizzler

The Rizzler

- If you wanted to include variables within text, format strings as per the following:

f-string Example

```
x = 3.14159
```

```
print(f'Approximate value of pi is {x}')
```

```
print(f'To three decimal places is {x:.3f}')
```

```
Approximate value of pi is 3.14159
```

```
To three decimal places is 3.142
```

- The string (surrounded by either single `' '` or double `" "` quotations) is preceded by `f`
- The variable is surrounded by curly braces `{}` within the string
- Formatting is applied to the variable *within the curly braces*
 - `A` : signals that the proceeding formatting code should be applied to the variable

f-string Formatting Basics

- `x:.3f`
 - Format variable `x`
 - Round to 3 decimal places
 - `f` means float

Section 5

Booleans

Booleans

- A boolean variable (type bool) can be either True (1) or False (0).

Booleans

```
x = True
y = 10
print(type(x), type(y))

print(y > 5)
print(y < 5)
print(type(y < 5))

<class 'bool'> <class 'int'>
True
False
<class 'bool'>
```

Booleans

```
x = 5
y = 10

print(x == 5)
print(x > 2 and y > 2)  # Both need to be True
print(x > 2 and y < 2)
print(x > 2 and not y < 2)
print(x > 2 or y < 2)  # At least one needs to be True
```

True

True

False

True

True

Comparison Operators

- > greater than
- < less than
- >= greater than or equal to
- <= less than or equal to
- == equal to
- != not equal to

Logical Operators

- and
- or
- not

Section 6

Conditional Statements

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Conditional Statements

- If thing1 happens, do x
- If thing2 happens instead, do y
- Otherwise, do z

Basic if/elif/else

```
x = 10 # Value presumably unknown  
y = 20
```

```
if x == 4:  
    print("x is 4")  
elif x > y:  
    print("x is bigger than y")  
else:  
    print("none of the above")
```

none of the above

Conditional Statements

Structure of Conditionals

```
if x == 4:  
    print("x is 4")
```

- Begins with `if/elif` ("else if")/else
- Boolean expression/value
- Colon :
- Indented code underneath that runs if condition is met

Section 7

Loops

while Loops

- Loops are used to run a chunk of code repeatedly
 - Often a set number of times
- This is best shown and unpacked through examples:

while Loops

while Loop Basics

```
count = 1    # Initialize counter
```

```
while count <= 5:
```

```
    print(count)
```

```
    count = count + 1    # Increment counter
```

1

2

3

4

5

while Loops

Iterating through a list

```
x = [0,1,2,3]
```

```
count = 0
```

```
while count <= len(x) - 1:
```

```
    print(x[count])
```

```
    count += 1      # Same as count = count + 1
```

0

1

2

3

while Loops

- The break statement is used to exit a loop

break

```
x = 0

while True:      # Infinite loop...
    if x > 10:
        break    # Until you break!
    x += 1

print(x)
```

11

for Loops

- Often used with the `range()` function to loop over code a set number of times

for Loops, `range()`

```
for i in range(3):  
    print(i)
```

0

1

2

for Loops

Iterating Over Entries in a List

```
x = [10,20,30,67]
```

```
for i in x:  
    print(i)
```

10

20

30

67

for Loops

- Use `enumerate()` and a second indexing variable (usually `i` or `j`) if you want to keep variable position

```
enumerate()
```

```
x = [10,20,30,67]
```

```
for i, j in enumerate(x):  
    print(i, j)
```

```
0 10
```

```
1 20
```

```
2 30
```

```
3 67
```

for Loops

- Loops (and conditionals) may be nested inside one another (like Russian dolls / Matryoshka)

Nesting

```
x = ["big", "small"]  
y = ["cat", "dog"]
```

```
for i in x:  
    for j in y:  
        print(i, j)
```

```
big cat  
big dog  
small cat  
small dog
```

Section 8

Importing Packages, Defining Functions

Importing Packages

- Some packages are installed with Python by default, and no extra work is required to use them

```
import math
```

```
import math
```

```
print(f'pi = {math.pi:.3f}')
```

```
pi = 3.142
```

Importing Packages

- We can “rename” packages to make their usage more concise

“Renaming” Packages

```
import math as m

print(f'pi = {m.pi:.3f}')
```



```
pi = 3.142
```

Importing Packages

- Some require installation. This is most easily (and safely) done through **anaconda**

Importing Packages

```
import numpy as np
```

```
import numpy as np          # Common convention
```

```
# Take a sample of 3 values from N(0, 1)
```

```
print(np.random.normal(loc = 0, scale = 1, size = 3))
```

```
[ 1.20404      1.37759701 -0.74875188]
```

Note

- $\text{loc} = \mu$, $\text{scale} = \sigma$

Defining Functions

- Sometimes you repeat the same code over and over again
 - When this happens, it is useful to collapse the repeated code into a function
 - Otherwise, code becomes long and messy due to copy-pasting

Defining Functions

Defining Basic Functions

```
def addition(a, b):  
    return a + b          # What the functions spits out  
  
print(addition(a = 1, b = 5)) # Call function  
  
def subtraction(a, b = 1):    # Give b default value of 1  
    return a - b  
  
print(subtraction(a = 5))
```

6

4

Defining Functions

Function Example

```
x = [0,1,2,3,4,5,6,7]
```

```
def qm_mean(things):  
    sum = 0  
  
    for i in things:  
        sum += i  
  
    return sum / len(things)
```

```
qm_mean(x)
```

3.5

Section 9

Numpy

- Somewhat similar to MATLAB if you have used that
- Useful for working with arrays/matrices
- Stands for **numerical python**
- Based in computer language **C**
 - Faster, more efficient computations

Working With Lists/Arrays

```
import numpy as np
```

```
list1 = [1,2,3,4]  
print(list1 * 2)  
print(type(list1))
```

```
numpyList = np.array(list1) # Convert list -> numpy array  
print(numpyList * 2)  
print(type(numpyList))
```

```
[1, 2, 3, 4, 1, 2, 3, 4]  
<class 'list'>  
[2 4 6 8]  
<class 'numpy.ndarray'>
```

- Matrices are typically worked with as **2D** numpy arrays

2D Numpy Arrays

```
import numpy as np

array2d = np.array([[1,2,3], [4,5,6]]) # List of lists
print(array2d)
print(type(array2d))

[[1 2 3]
 [4 5 6]]
<class 'numpy.ndarray'>
```


To Check Attributes of a Numpy Variable

```
import numpy as np

array2d = np.array([[1,2,3], [4,5,6]])

print(array2d.shape)      # Shape (# rows, # columns)
print(array2d.ndim)       # Number of dimensions
print(array2d.size)       # Number of elements
print(array2d.dtype)      # Type of elements

(2, 3)
2
6
int64
```

- Some numpy types:
 - `uint8`
 - `int32`
 - `int64`
 - `float64`
- Generally, numpy type = type learned before + memory (in bits)

```
np.arange()
```

```
import numpy as np
```

```
np.arange(start=10, stop=20, step=2, dtype = "int64")
```

```
array([10, 12, 14, 16, 18])
```

Note

- stop parameter is *exclusive*

```
np.linspace()
```

```
import numpy as np
```

```
np.linspace(start=10, stop=20, num=6, dtype = "int64")
```

```
array([10, 12, 14, 16, 18, 20])
```

Tip

- Use `np.linspace()` for breaking something into equal parts
- Use `np.arange()` for creating a sequence of numbers

- Numpy has lots of functions relating to choosing random values

Random Choice

```
import numpy as np

print(np.random.choice(["A", "B", "C", "D"]))

print(np.random.choice(["A", "B", "C", "D"], size = 2))
```

A

['B' 'B']

Random Numbers

```
import numpy as np

print(np.random.normal(loc = 0, scale = 1, size = 3))
print(np.random.uniform(low = 0, high = 10, size = 3))
```

```
[-0.71577323 -1.16307701 -0.29448465]
[4.94375595 3.93611213 9.94769751]
```

Indexing Numpy Arrays

```
import numpy as np

x = np.random.normal(0, 10, size=(3,3))
print(x, "\n")  # Print x with a blank line after it

print(x[2,2], type(x[2,2]))
print(x[2,:])    # Row 2 (zero-based), all columns

[[ 14.31658315 -4.9705413 -11.72691219]
 [-16.92988425 -1.73309841  4.90880598]
 [ 21.88149086 -7.56058598 -17.11192795]]

-17.111927946287015 <class 'numpy.float64'>
[ 21.88149086 -7.56058598 -17.11192795]
```

Indexing (cont.)

```
import numpy as np

x = np.random.normal(0, 10, size=(3,3))
print(x, "\n")

print(x[2, 0:2])      # Row 2, columns 0 and 1
print(type(x[2, 0:2]))
```

```
[[ -9.52649203   1.21612739  -5.94365974]
 [  1.5301394  -16.08046256   3.18125409]
 [ 16.09322135   6.46190693   9.9946063  ]]
```

```
[16.09322135  6.46190693]
```

```
<class 'numpy.ndarray'>
```


- You may also perform **aggregate** functions on numpy arrays

Numpy Aggregate Functions

```
import numpy as np

x = np.array([1,4,7,15])

print(np.mean(x))
print(np.median(x))
print(np.std(x))
```

6.75

5.5

5.2141634036535525

More Aggregate Functions

```
import numpy as np

x = np.array([1,4,7,15])

print(np.min(x))           # Minimum value
print(np.argmin(x))        # Position of minimum value
print(np.max(x))           # Maximum value
print(np.argmax(x))        # Position of maximum value
```



```
1
0
15
3
```

Check-in

- How might you generate 100 random values from the standard normal distribution?
- How might you get the maximum of these values?

Check-in

- How might you generate 100 random values from the standard normal distribution?
- How might you get the maximum of these values?

```
import numpy as np
```

```
print(np.max(np.random.normal(loc=0, scale=1, size=100)))
```

```
2.1174076641575197
```

Section 10

Pandas

- For data analysis in Python
- Somewhat similar to R if you have used that
 - Introduces the data frame

Creating a DataFrame

```
import pandas as pd

pd.DataFrame({
    "Memes" : ["67", "the rizzler", "harambe"],
    "Rating": [5, 9, 10]
})
```

	Memes	Rating
0	67	5
1	the rizzler	9
2	harambe	10

Section 11

Matplotlib.pyplot

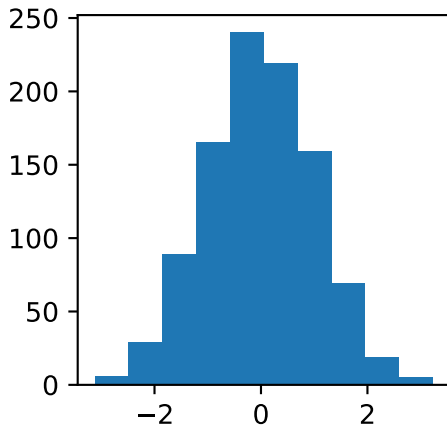
- **Matplotlib.pyplot** is a package for visualizing data in Python
- Fun fact: If you have asked ChatGPT for a graph, you likely received one that was made using this package

Basic Histogram

```
import numpy as np
import matplotlib.pyplot as plt

x = np.random.normal(0, 1, 1000)
plt.hist(x)
plt.show()
```

Basic Histogram

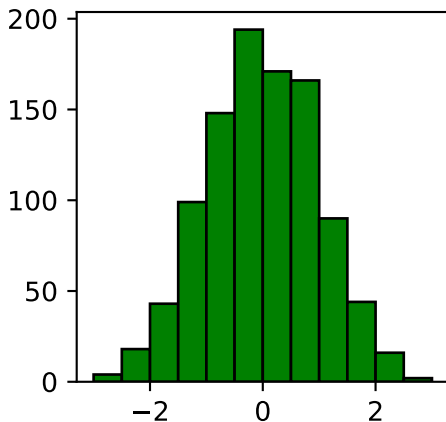


- A few ways to modify the histogram are illustrated below

Modifying the Plot

```
plt.hist(  
    x,                                # Data  
    color = "green",                 # Bar Colour  
    edgecolor = "black",             # Outline Colour  
    bins = np.arange(-3, 3.5, 0.5) # Bins of 0.5 from -3 to +3  
)  
plt.show()
```

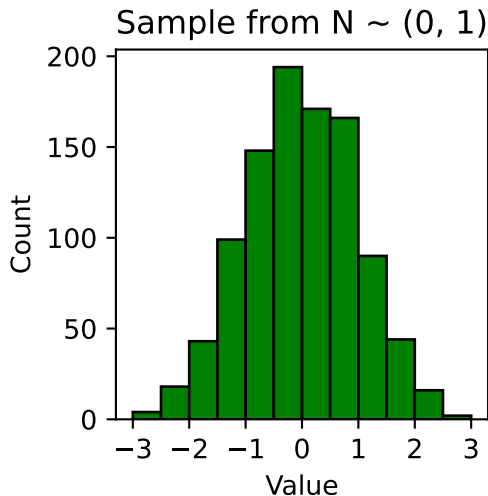
Modifying the Plot



More Ways to Modify

```
plt.hist(  
    x, color = "green", edgecolor = "black",  
    bins = np.arange(-3, 3.5, 0.5)  
)  
  
plt.xticks(np.arange(-3, 4, 1)) # x-axis ticks spaced by 1  
plt.title("Sample from  $N \sim (0, 1)$ ")  
plt.xlabel("Value")  
plt.ylabel("Count")  
plt.show()
```

More Ways to Modify



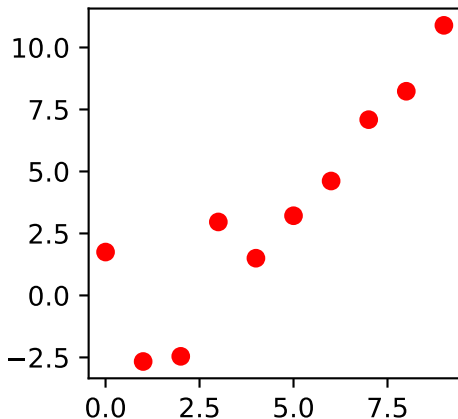
- The below is how data was generated for subsequent scatterplot examples

Simulated Data for a Simple Regression

```
errors = np.random.normal(0, 2, 10) # Normal errors
nums_x = np.arange(0, 10, 1) # Nums 0 to 9
nums_y = nums_x + errors #  $y = x + \text{errors}$ 
```


Scatterplot

```
plt.plot(nums_x, nums_y, 'ro') # plot(x, y, style)
plt.show()                     # 'ro' = red circles
```



Section 12

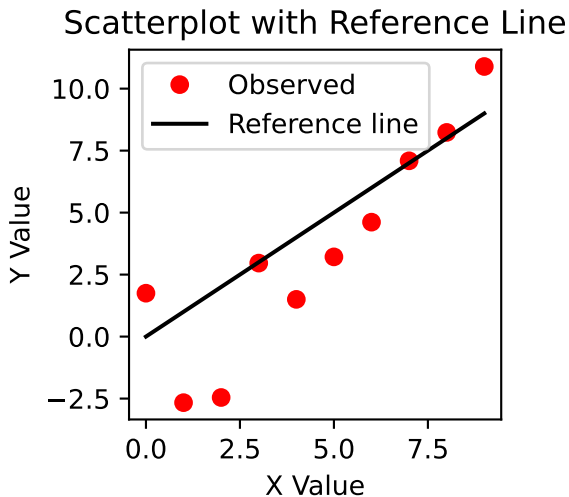
Matplotlib.pyplot

- You can add another plot layer by calling another plotting function before `plt.show()`
- This is useful for adding:
 - reference lines
 - fitted lines
 - multiple groups

Adding a Line to a Scatterplot

```
plt.plot(nums_x, nums_y, 'ro')      # Observed data: red circles
plt.plot(nums_x, nums_x, 'k-')      # Reference line: black solid line
plt.show()
```

Adding Labels and a Legend



Section 13

statsmodels

- statsmodels is a package for statistical testing and modelling
- It is useful when you want:
 - formal hypothesis tests
 - regression models
 - p-values and confidence intervals
 - detailed statistical output

Importing statsmodels Functions

```
import statsmodels.formula.api as smf
from statsmodels.stats.weightstats import ttest_ind
```

- `statsmodels.formula.api` contains functions that use formula notation
- It is commonly renamed `smf`
- `ttest_ind` is a function for independent-samples t-tests

Import Structure

```
import statsmodels.formula.api as smf
#      package.module          alias

from statsmodels.stats.weightstats import ttest_ind
#      package.module.submodule      function
```

- Statistical tests help us use a **sample** to make inferences about a larger population
- A test usually starts with a **null hypothesis**
 - Often “no difference” or “no relationship”
- We compare the observed data to what would be expected if the null hypothesis were true

Note

- The code runs the test, but the researcher chooses whether the test makes sense

General Workflow

- 1 State the research question
- 2 Identify the variable types and study design
- 3 Summarize and visualize the data
- 4 Run the appropriate test
- 5 Interpret the estimate, confidence interval, and p-value

Example Data

- We will use simulated scores from two independent groups
- Each row represents one participant
- The group labels are stored in a pandas DataFrame

Creating the Data

```
import pandas as pd

np.random.seed(42)

scores_df = pd.DataFrame({
    "group": ["control"] * 30 + ["treatment"] * 30,
    "score": np.concatenate([
        np.random.normal(70, 10, 30),
        np.random.normal(76, 10, 30)
    ])
})
```

Inspecting the Data

```
print(scores_df.head())    # First five rows
print(scores_df.shape)     # (number of rows, number of columns)
```

	group	score
0	control	74.967142
1	control	68.617357
2	control	76.476885
3	control	85.230299
4	control	67.658466

(60, 2)

- Always inspect the data before running a test

Example Data

Group Summary

```
print(  
    scores_df.groupby("group").agg(  
        n = ("score", "count"),  
        mean = ("score", "mean"),  
        sd = ("score", "std")  
    )  
)
```

	n	mean	sd
group			
control	30	68.118531	9.000064
treatment	30	74.788375	9.311022

- `groupby("group")` summarizes the control and treatment rows separately

Independent-Samples t-Test

- Used to compare the means of **two independent groups**
- Example: Do treatment and control groups have different mean scores?
- Do not use this test if the same participants appear in both groups

Null hypothesis:

$$H_0 : \mu_{\text{treatment}} - \mu_{\text{control}} = 0$$

Independent-Samples t-Test

Selecting Each Group

```
control = scores_df.loc[
    scores_df["group"] == "control", # Rows to include
    "score"                          # Column to return
]

treatment = scores_df.loc[
    scores_df["group"] == "treatment",
    "score"
]
```

- `ttest_ind()` needs the two groups as separate variables

Welch's t-Test

- Welch's t-test is an independent-samples t-test
- It does **not** assume equal population variances
- It is a useful default when comparing two independent groups

Running the Test

```
test_result = ttest_ind(  
    treatment,      # First group  
    control,        # Second group  
    usevar = "unequal"  
)
```

```
print(test_result)
```

```
(np.float64(2.82107476851771), np.float64(0.006544738502602257))
```

Function Structure

```
ttest_ind(  
    x1,                # Values from the first group  
    x2,                # Values from the second group  
    alternative = "two-sided",  
    usevar = "unequal", # Welch's t-test  
    value = 0           # Difference under the null hypothesis  
)
```

- Use `alternative = "two-sided"` when either group could have the larger mean
- Use `value = 0` when the null hypothesis is no mean difference

ttest_ind()

- ttest_ind() returns three values:

- 1 t-statistic
- 2 p-value
- 3 degrees of freedom

Tuple Unpacking

```
t_statistic, p_value, degrees_freedom = ttest_ind(
    treatment,
    control,
    usevar = "unequal"
)
```

```
print(t_statistic)
print(p_value)
print(degrees_freedom)
```

2.82107476851771

t-Test Interpretation

- The t-statistic describes the group difference relative to its uncertainty
- The p-value describes how incompatible the result is with the null hypothesis
- The degrees of freedom help determine the t-distribution used by the test

! Important

- A p-value is **not** the probability that the null hypothesis is true

Mean Difference

- The t-test output should be interpreted alongside the observed group difference

Calculating the Mean Difference

```
mean_difference = treatment.mean() - control.mean()

print(f"Treatment mean: {treatment.mean():.2f}")
print(f"Control mean: {control.mean():.2f}")
print(f"Mean difference: {mean_difference:.2f}")
```

Treatment mean: 74.79

Control mean: 68.12

Mean difference: 6.67

Reporting a t-Test

Formatting the Result

```
print(  
    f"Mean difference = {mean_difference:.2f}, "  
    f"t({degrees_freedom:.2f}) = {t_statistic:.2f}, "  
    f"p = {p_value:.3f}"  
)
```

Mean difference = 6.67, t(57.93) = 2.82, p = 0.007

- Report the observed difference and p-value together
- Use cautious language such as “evidence of a difference”

- A correlation describes the direction and strength of a **linear relationship**
- Pearson's correlation coefficient is represented by r
- It ranges from -1 to 1

! Important

- Correlation does not establish causation

Adding a Second Variable

```
np.random.seed(7)

scores_df["study_hours"] = (
    2
    + 0.08 * scores_df["score"]
    + np.random.normal(0, 1.5, len(scores_df))
)
```

Scatterplot

```
plt.plot(scores_df["study_hours"], scores_df["score"], "bo")  
plt.xlabel("Study Hours")  
plt.ylabel("Score")  
plt.show()
```

- Use a scatterplot before calculating a correlation

`.corr()` Method

```
correlation = scores_df["study_hours"].corr(  
    scores_df["score"]  
)
```

```
print(f"r = {correlation:.2f}")
```

```
r = 0.53
```

- `.corr()` describes the observed sample relationship
- Regression can test whether the population slope differs from zero

Simple Linear Regression

- Simple linear regression examines the relationship between:
 - one continuous **outcome**
 - one continuous **predictor**
- Here, the outcome is `score` and the predictor is `study_hours`

$$\text{score} = \text{intercept} + (\text{slope} \times \text{study hours}) + \text{error}$$

Formula Structure

```
"score ~ study_hours"  
# outcome ~ predictor
```

- The outcome goes on the left of ~
- The predictor goes on the right of ~
- Column names must match the DataFrame

Creating a Model

```
model_definition = smf.ols(  
    formula = "score ~ study_hours",  
    data = scores_df  
)
```

```
print(type(model_definition))
```

```
<class 'statsmodels.regression.linear_model.OLS'>
```

- ols stands for **ordinary least squares**
- smf.ols() defines the model but does not estimate its coefficients yet

Fitting a Model

```
model = model_definition.fit()
```

```
print(type(model))
```

```
<class 'statsmodels.regression.linear_model.RegressionResultsWrapper'>
```

- .fit() estimates the intercept, slope, and other model statistics
- Store the fitted result so its attributes can be inspected

Defining and Fitting in One Step

```
model = smf.ols(  
    "score ~ study_hours",  
    data = scores_df  
)
```

- This is a common way to define and fit a model

Model Parameters

```
print("Intercept:", round(model.params["Intercept"], 2))  
print("study_hours:", round(model.params["study_hours"], 2))
```

Intercept: 50.43
study_hours: 2.75

- Intercept: predicted score when study hours equals zero
- study_hours: predicted score change for one additional study hour

Accessing One Parameter

```
intercept = model.params["Intercept"]  
slope = model.params["study_hours"]  
  
print(f"Intercept: {intercept:.2f}")  
print(f"Slope: {slope:.2f}")
```

Intercept: 50.43

Slope: 2.75

- Use the parameter name to extract a specific estimate

Testing the Slope

```
slope_p_value = model.pvalues["study_hours"]  
  
print(f"study_hours p-value: {slope_p_value:.3f}")
```

study_hours p-value: 0.000

Null hypothesis:

$$H_0 : \text{population slope} = 0$$

`.conf_int()`

Confidence Intervals

```
slope_ci = model.conf_int().loc["study_hours"]  
  
print(f"Lower: {slope_ci[0]:.2f}")  
print(f"Upper: {slope_ci[1]:.2f}")
```

Lower: 1.59

Upper: 3.90

- A confidence interval gives a range of values compatible with the model and data

```
.summary().tables[0]
```

Model Overview

```
summary = model.summary()  
print(summary.tables[0])
```

```
=====
                        OLS Regression Results
=====
Dep. Variable:          score      R-squared:          0.280
Model:                  OLS        Adj. R-squared:       0.268
Method:                 Least Squares  F-statistic:       22.57
Date:                  Mon, 08 Jun 2026  Prob (F-statistic): 1.37e-05
Time:                  15:49:36      Log-Likelihood:    -210.99
No. Observations:      60           AIC:               426.0
Df Residuals:          58           BIC:               430.2
Df Model:               1
Covariance Type:       nonrobust
=====
```

```
.summary().tables[1]
```

Coefficient Table

```
print(summary.tables[1])
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	50.4278	4.553	11.076	0.000	41.314	59.541
study_hours	2.7466	0.578	4.751	0.000	1.589	3.904

- Table 1 gives the estimates for each model term
- Focus first on coef, $P>|t|$, and [0.025, 0.975]

```
.summary().tables[2]
```

Diagnostics Table

```
print(summary.tables[2])
```

```
=====
Omnibus:                0.940    Durbin-Watson:                1.864
Prob(Omnibus):           0.625    Jarque-Bera (JB):          0.843
Skew:                   -0.281    Prob(JB):                  0.656
Kurtosis:               2.852    Cond. No.                  34.0
=====
```

- Table 2 contains diagnostic and assumption-related information
- Useful later when checking whether the model is behaving reasonably

.rsquared

- R-squared is the proportion of outcome variance described by the fitted model
- It ranges from 0 to 1
- A larger R-squared does not prove that the model is correct or causal

R-Squared Attribute

```
print(f"R-squared: {model.rsquared:.3f}")
```

```
R-squared: 0.280
```

.predict()

Predicted Outcomes

```
scores_df["predicted_score"] = model.predict(scores_df)

for i in range(3):
    print(
        f"Observed: {scores_df['score'][i]:.2f}; "
        f"Predicted: {scores_df['predicted_score'][i]:.2f}"
    )
```

Observed: 74.97; Predicted: 79.36

Observed: 68.62; Predicted: 69.08

Observed: 76.48; Predicted: 72.86

- .predict() calculates the fitted outcome for each row

Plotting Model Predictions

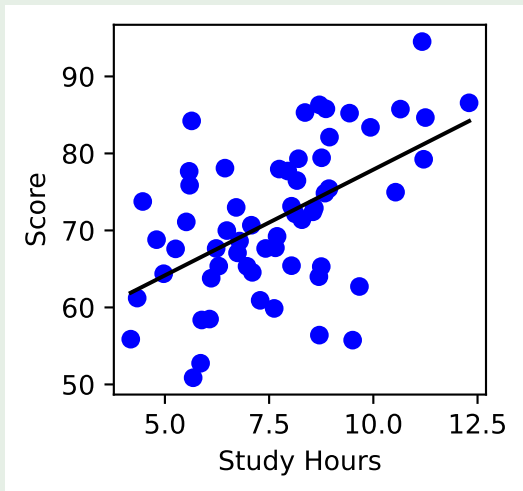
Observed and Predicted Values

```
ordered = scores_df.sort_values("study_hours")

plt.plot(scores_df["study_hours"], scores_df["score"], "bo")
plt.plot(ordered["study_hours"], ordered["predicted_score"], 'b-')
plt.xlabel("Study Hours")
plt.ylabel("Score")
plt.show()
```

Plotting Model Predictions

Fitted Line



- Blue circles are observed values

Reporting Regression Results

Formatting the Result

```
slope_ci = model.conf_int().loc["study_hours"]

print(
    f"b = {model.params['study_hours']:.2f}, "
    f"95% CI [{slope_ci[0]:.2f}, {slope_ci[1]:.2f}], "
    f"p = {model.pvalues['study_hours']:.3f}"
)
```

b = 2.75, 95% CI [1.59, 3.90], p = 0.000

- Report the estimated slope, confidence interval, and p-value together
- Use **associated with**, rather than **caused**